Matrix Interpretation of the DFT

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Key Concepts

1) The length-N DFT can be described as a matrix multiplication.

    a) The time samples are arranged in a length-N column vector $\mathbf{x}$.

    b) The DFT coefficients are arranged in a length-N column vector $\mathbf{X}$.

    c) We write $\mathbf{X} = \mathbf{W}\mathbf{X}$ where W is the N-by$N$ DFT matrix.

    d) The m, p element of $\mathbf{W}$ is $e^{-j(m-1)\frac{2\pi}{N}(p-1)}$ where $k = m - 1$ is the DFT frequency index and $n = p - 1$ is the DFT time index.

2) The inverse DFT is $\mathbf{W}^{-1}$, the matrix inverse of $\mathbf{W}$.

3) The inverse DFT reveals that the vector of time samples $\mathbf{x}$ is expressed as a weighted sum of complex sinusoidal basis vectors with frequencies an integer multiple of $\frac{2\pi}{N}$.

Video Screenshots

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The DFT maps N numbers - $x[0], x[1], \ldots x[N-1]$ - into N new numbers - $X[0], X[1], \ldots, X[N-1]$

Interpret this operation as the product of an $N \times N$ DFT matrix and an $N \times 1$ vector of signal values.

- useful way to view DFT
- power of matrix (linear) algebra in signal processing

Example ($N = 4$):

$$X[k] = \sum_{n=0}^{3} x[n]\, e^{-j k \frac{2\pi}{4} n}$$

$$= x[0]\, W_4^{k0} + x[1] W_4^{k1} + x[2] W_4^{k2} + x[3] W_4^{3k}$$

$$W_N = e^{-j 2\pi/N}$$

$$\begin{matrix} k=0 \\ k=1 \\ k=2 \\ k=3 \end{matrix} \begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

Let $\underline{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}$, and $\underline{X} = \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}$ $\Rightarrow \underline{X} = \underline{W}\,\underline{x}$

where $\underline{W} = \begin{bmatrix} W_N^0 & W_N^0 & W_N^0 & \cdots & W_N^0 \\ W_N^0 & W_N^1 & W_N^2 & \cdots & W_N^{N-1} \\ W_N^0 & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & & & & \\ W_N^0 & W_N^{N-1} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)^2} \end{bmatrix}$

$$X[k] = W_N^0 x[0] + W_N^k x[1] + W_N^{2k} x[2] + \cdots + W_N^{k(N-1)} x[N-1]$$

The Inverse DFT

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If $\underline{X} = \underline{W}\,\underline{x}$ is the DFT, then $\underline{x} = \underline{W}^{-1}\underline{X}$ is the inverse DFT

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\, e^{j k \frac{2\pi}{N} n} = \frac{1}{N}\left(X[0]W_N^{-0} + X[1]W_N^{-n} + X[2]W_N^{-2n} + \cdots + X[N-1]W_N^{-(N-1)n} \right)$$

$$\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix} = \frac{1}{N} \underbrace{\begin{bmatrix} W_N^{-0} & W_N^{-0} & W_N^{-0} & \cdots & W_N^{-0} \\ W_N^{-0} & W_N^{-1} & W_N^{-2} & \cdots & W_N^{-(N-1)} \\ W_N^{-0} & W_N^{-2} & W_N^{-4} & \cdots & W_N^{-2(N-1)} \\ \vdots & \vdots & \vdots & & \vdots \\ W_N^{-0} & W_N^{-(N-1)} & W_N^{-2(N-1)} & \cdots & W_N^{-(N-1)^2} \end{bmatrix}}_{\underline{W}^{-1}} \begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ X[N-1] \end{bmatrix}$$

complex conjugate transpose

Note: $\underline{W}^{-1} = \frac{1}{N}\underline{W}^{H}$

Check: $\underline{W}\,\underline{W}^{-1} = \underline{W}\,\frac{1}{N}\underline{W}^{H} = \underline{I}$.

DFT Basis Vectors

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Let $\underline{w}_\ell = [\underline{W}^{-1}]_{:,\ell}$ (ℓ^{th} column of $\underline{W}^{-1}$)

then $\underline{x} = \underline{W}^{-1}\underline{X} \Rightarrow \underline{x} = \underline{w}_1 X[0] + \underline{w}_2 X[1] + \underline{w}_3 X[2] + \cdots + \underline{w}_N X[N-1]$

$\underline{w}_1, \underline{w}_2, \ldots \underline{w}_N$ are <u>basis vectors</u> for representing $\underline{x}$

$$\underline{w}_\ell = \frac{1}{N}\begin{bmatrix} 1 \\ e^{j\ell\frac{2\pi}{N}} \\ e^{j\ell\frac{2\pi}{N}2} \\ \vdots \\ e^{j\ell\frac{2\pi}{N}(N-1)} \end{bmatrix}$$

complex sinusoid with frequency $\ell\frac{2\pi}{N}$

Matrix-Vector Versions of DFT properties

Ex: Parseval

$$\frac{1}{N}\sum_{k=0}^{N-1}|X[k]|^2 = \frac{1}{N}\underline{X}^H\underline{X} \quad ; \ \underline{X} = \underline{W}\underline{x}$$

$$= \frac{1}{N}\underline{x}^H\underline{W}^H\underline{W}\underline{x} \quad ; \ \underline{W}^H\underline{W} = N\underline{I}$$

$$= \underline{x}^H\underline{x}$$

$$= \sum_{n=0}^{N-1}|x[n]|^2$$

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